Based on the provided sources, here is an explanation of private and public IP addressing:

1. Definitions

- **Private IP Address (Local Address – LA):** A private IP address is used exclusively inside a local network and is **not visible on the Internet** 1. It serves to identify a specific system or device within a local environment, such as a computer lab 2.
- **Public IP Address:** A public IP address uniquely identifies a network or an individual device on the **Internet** 1, 3. It acts as the global identifier for an entire local network when communicating with the outside world 2.

2. Differences Between Private and Public IP Addresses

- **Visibility:** Private IPs are only meaningful and visible within the internal network (like a college or office), whereas Public IPs are visible to everyone on the Internet 1, 3.
- **Uniqueness:** A Public IP must be globally unique to identify a network on the Internet, while Private IPs only need to be unique within their specific local network 1, 2.
- **Scope:** Private IPs are logical addresses assigned by the network to identify systems within a lab, while Public IPs represent that entire lab to the rest of the world 2.

3. Advantages of Private Addressing

- **Internal Organization:** They allow for the structured identification of devices within a local area without exhausting globally unique addresses 1, 2.
- **Privacy and Security:** Because they are "not visible on the Internet," they provide a layer of isolation for internal devices 1.
- **Resource Management:** In virtualization, private IPs can be assigned during the **provisioning** stage of the VM lifecycle to configure internal network settings for new virtual machines 4.

4. Real-World Usage Examples

- **The College Analogy:** Inside a college, every student has a **roll number** (e.g., Roll No: 21). This roll number is like a **Private IP** because it is meaningful only inside that college 1. However, the college has one **official postal address** (e.g., TKM College of Engineering, Kollam). This postal address is the **Public IP**, as people outside the college use it to find the institution but do not know individual student roll numbers 3.
- **Computer Lab:** In a typical lab setup, the **MAC address** identifies the hardware, a **Private IP** identifies the specific computer within the lab network, and a **Public IP** represents the entire lab on the Internet 2.

5. Exam-Ready Comparison Table

Feature,Private IP Address,Public IP Address
Scope,Local network only 1.,Global Internet 3.
Visibility,Not visible on the Internet 1.,Visible to the entire Internet 1.

Analogy,Student Roll Number 1.,College Postal Address 3.
Function,Identifies systems within a lab 2.,Identifies a network on the Internet 2.
Assignment,"Assigned by local network/Hypervisor 2, 5.",Assigned to identify the network globally 2.

Note: *The provided sources explain the concepts and analogies of IP addressing but do not explicitly list the specific numerical* ***address ranges*** *(such as the RFC 1918 ranges like 192.168.x.x). You may wish to independently verify those standard ranges for a complete technical answer.*